Seeking Interpretability and Explainability in Binary Activated Neural Networks

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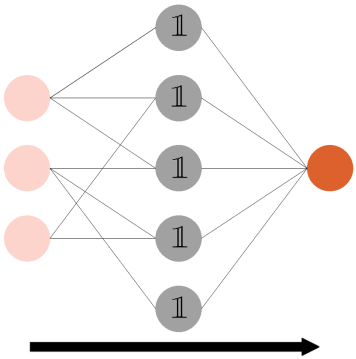
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Simple neural networks

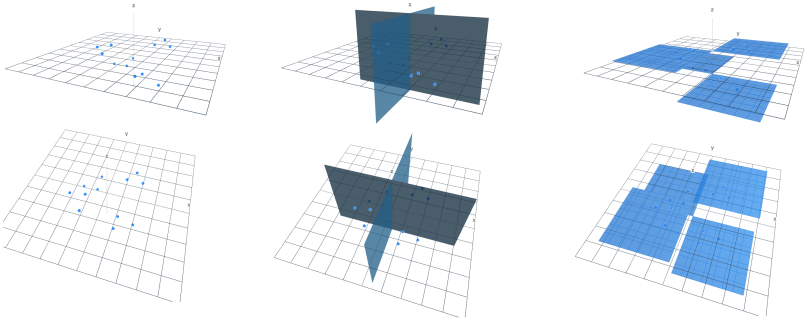
- Feedforward neural network
- Binary activations (threshold)

$$\mathbb{1}_{\{x\}} = \begin{cases} 1 & \text{if } x \geq 0, \\ 0 & \text{otherwise.} \end{cases}$$

- Shallow networks (1 hidden layer)
- Narrow networks
- Sparse (2)

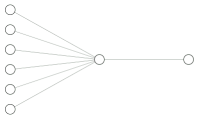


Understanding binary activated neural networks



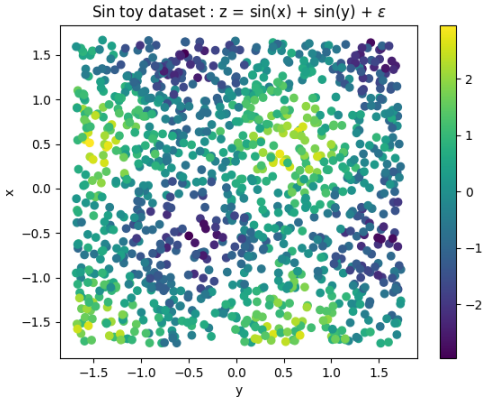
Characteristics

- For tackling regression tasks
- Greedy algorithm (inspired from Adaboost)
- Uses a L1 norm regularization
- Convergence guarantees on the train mean squared error
- No fixed architecture
- Fewer hyperparameter (no learning rate, batch size, ...)

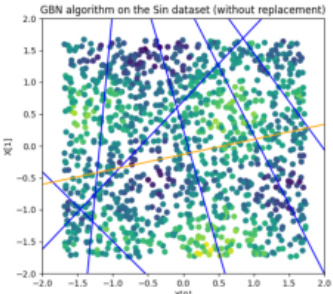


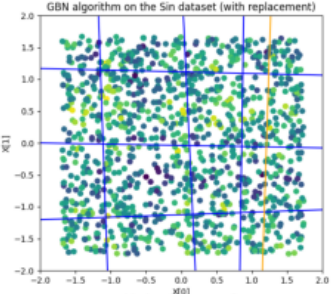


Tackling the drawbacks of greediness



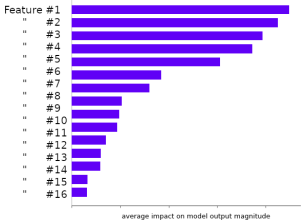
Tackling the drawbacks of greediness





SHAP values

- Goal: quantifying the contribution (magnitude, impact) of each feature to a prediction
- At a prediction or a dataset level of aggregation



Conclusion

Artificial neural networks can be **interpretable** predictors...
... When trained with such a goal in mind!

References

[1] J. Lin, C. Gan, and S. Han, "Defensive quantization: When efficiency meets robustness," CoRR, vol. abs/1904.08444, 2019.

[2] M. Courbariaux, I. Hubara, D. Soudry, R. El-Yaniv, and Y. Bengio, "Binarized neural networks: Training deep neural networks with weights and activations constrained to +1 or -1," 2016.

Thank you for your attention :)

